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CogMem

Cognitively-Grounded Long-Term Memory for Conversational Agents

**Typed memory graph + multi-channel recall
+ SUM spreading activation**

Defense focus

- What was built: a typed memory graph pipeline.
- What was proven: judge-based LongMemEval and LoCoMo results.
- What remains: temporal errors and habit-specific validation.



10-15 minute storyline:
problem -> HINDSIGHT -> method ->
retain demo -> experiments -> proof

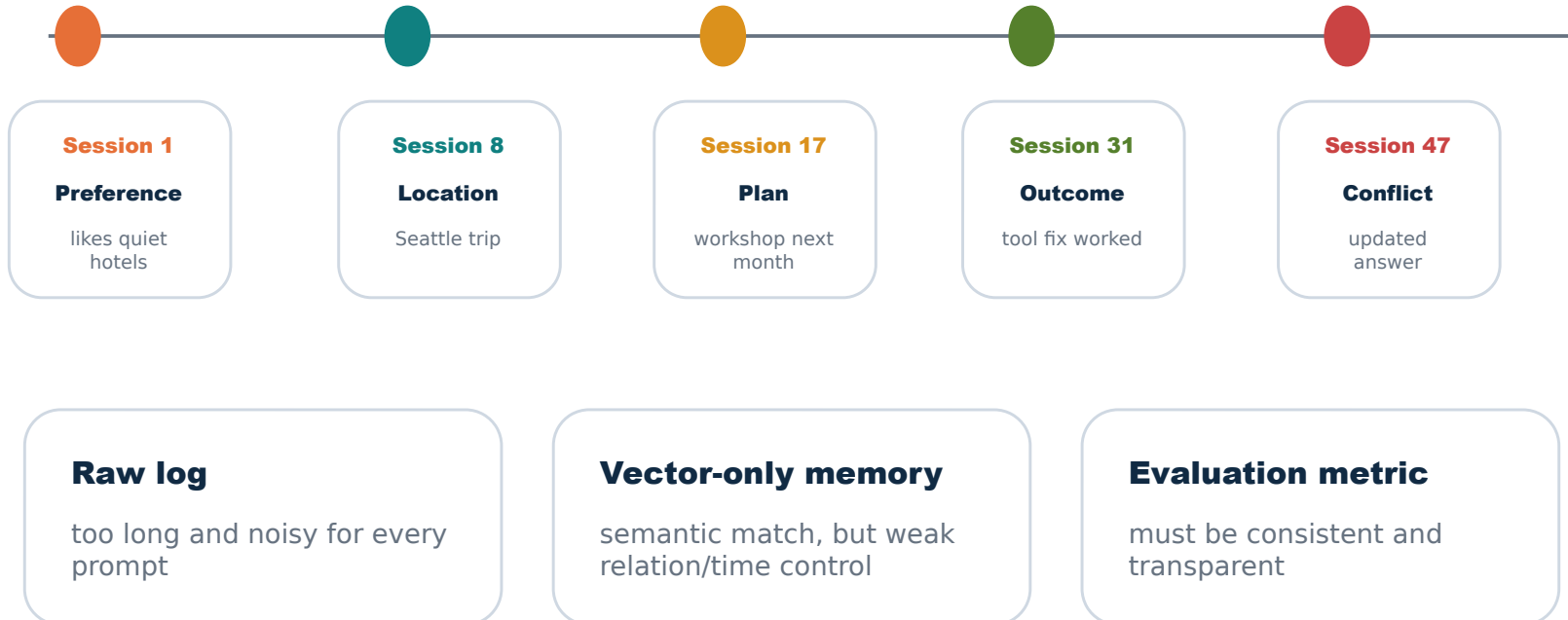
Student: Le Minh Triet

ONE LOVE. ONE FUTURE.

Problem: long conversations break short-context memory

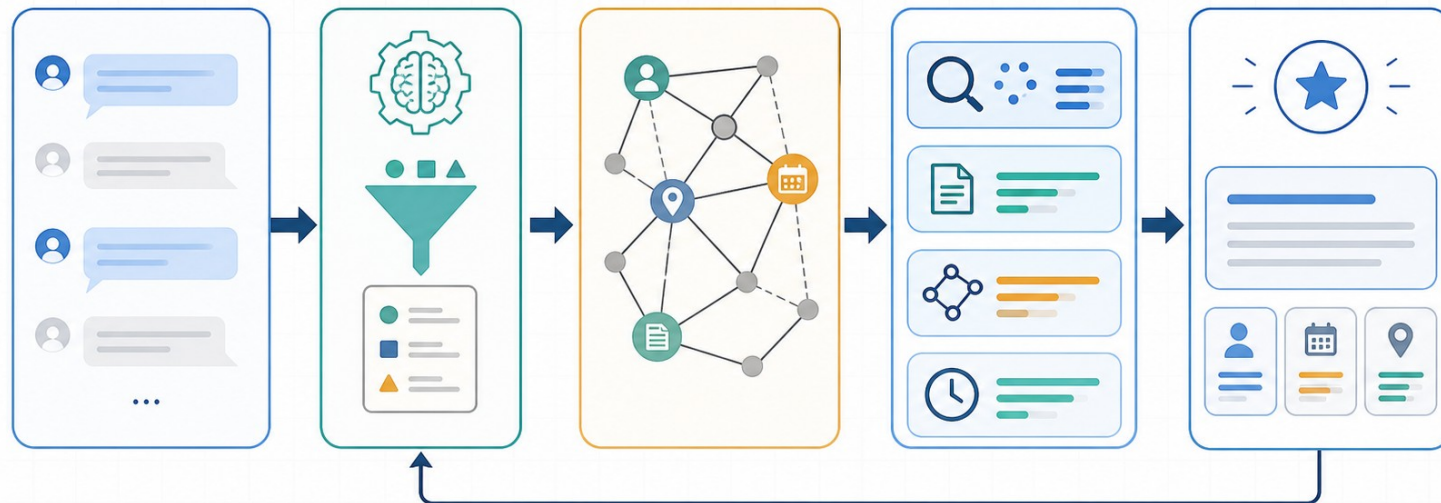
MOTIVATION

A useful assistant must remember across many sessions, but the answer is rarely stored as one clean sentence.



HINDSIGHT starts from observations before graph recall

BASELINE



1. Dialogue chunk

2. Observation
extraction

3. Memory graph

4. Retrieval + answer

Why CogMem changes it: fewer node types are not enough; the retain path must be faster, more typed, and less node-heavy.

Retain walkthrough: raw dialogue becomes typed evidence

MEMORY REPRESENTATION

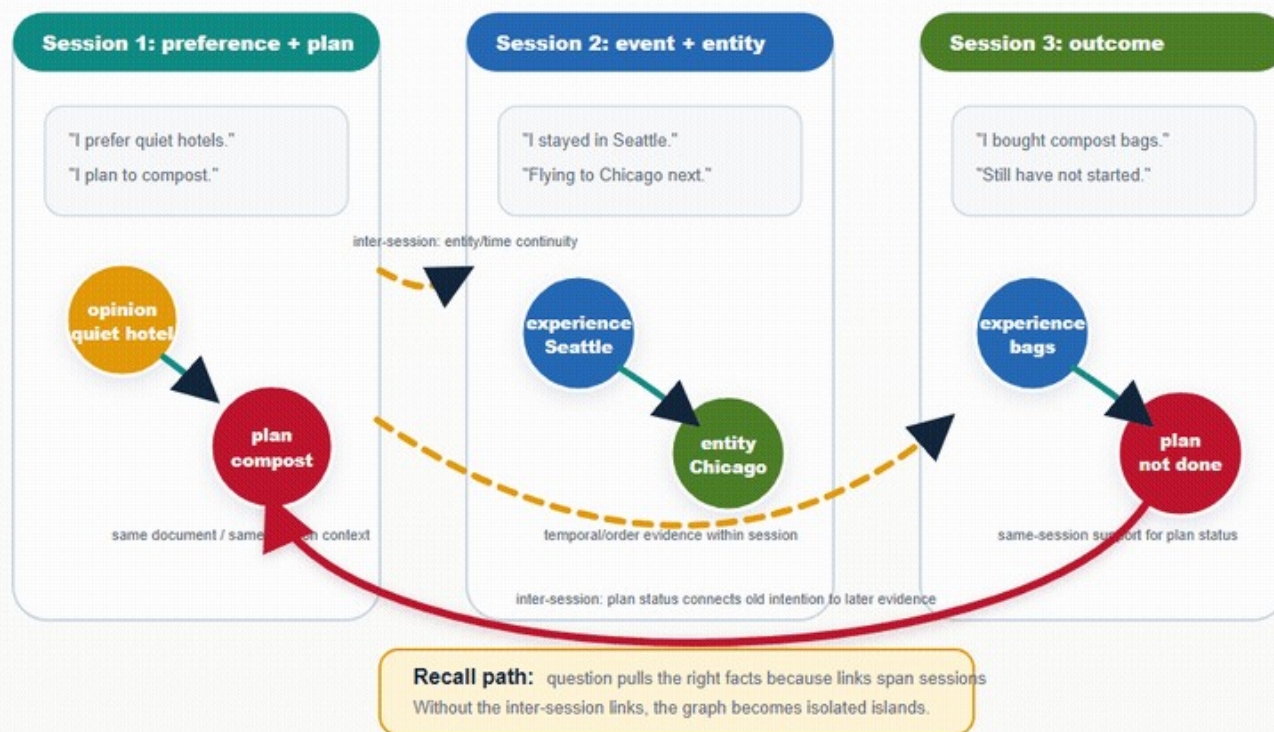
Retain builds typed facts, then links them inside and across sessions

The important part is not only extraction: CogMem also creates intra-session and inter-session graph structure over time.

Retain pipeline

1. Read raw dialogue
2. Extract typed facts
3. Store nodes + metadata
4. Add graph links

- intra-session link
- - inter-session link
- recalled path



Animation: solid teal = intra-session links; dashed gold = inter-session links; red = recalled evidence path.

Six typed memory networks

MEMORY REPRESENTATION



World

stable facts

Experience

dated events

Opinion

preferences

Habit

routines

Intention

future plans

Action-effect

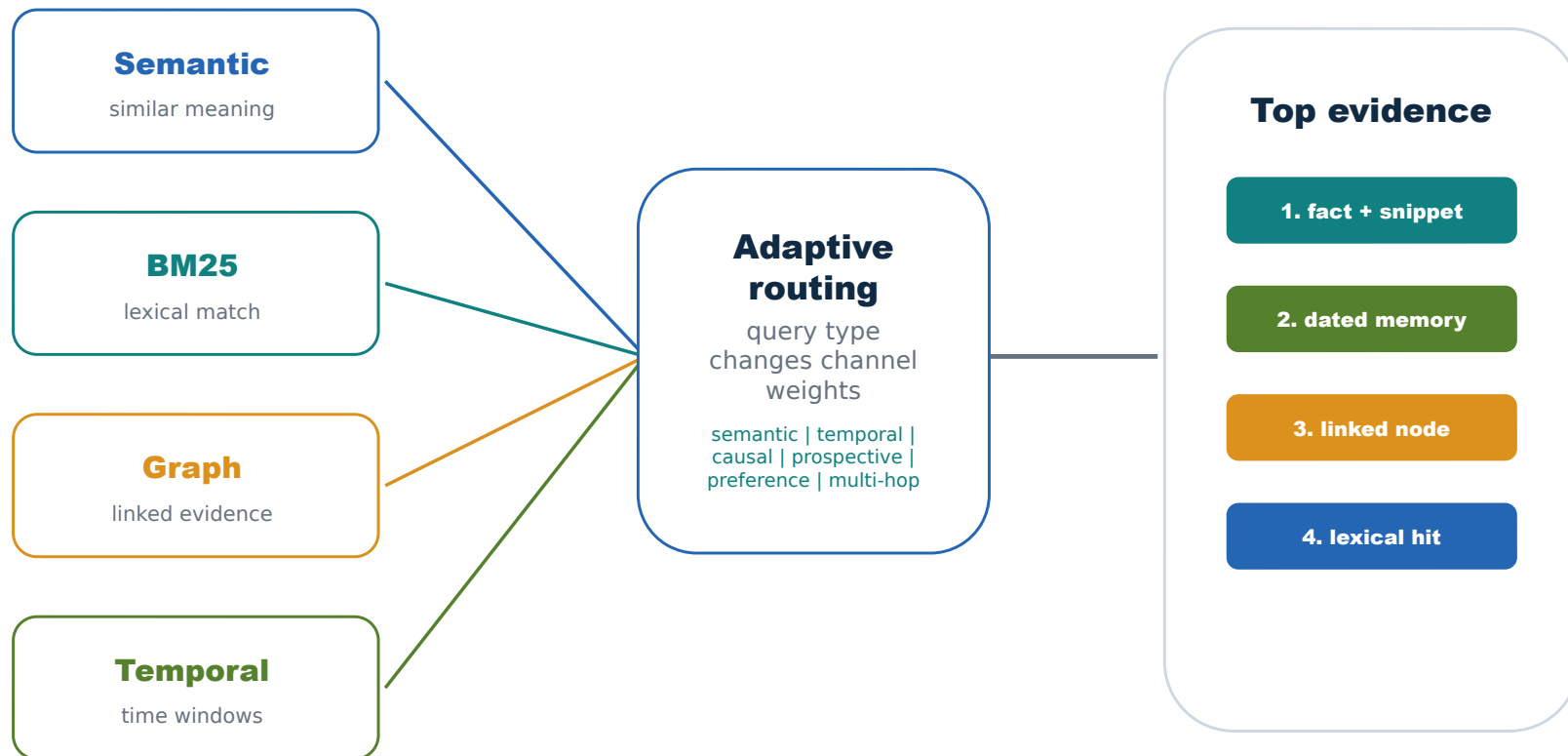
precondition -> action
-> outcome

Honest scope

Intention and action-effect have targeted qualitative ablations. Habit is representationally useful, but still needs a routine diary workload.

Recall: four channels become one ranked evidence list

METHOD



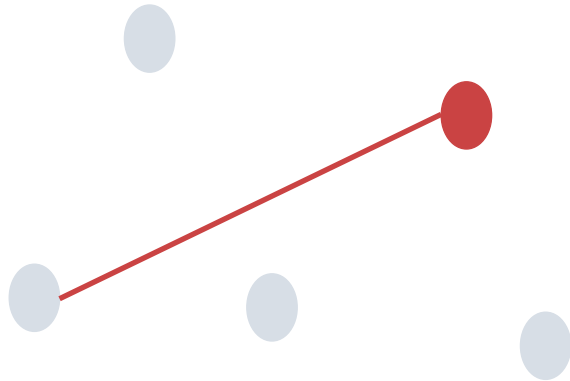
Key claim: graph is important, but the final system is deliberately multi-channel.

Graph recall: SUM instead of MAX

METHOD

MAX baseline

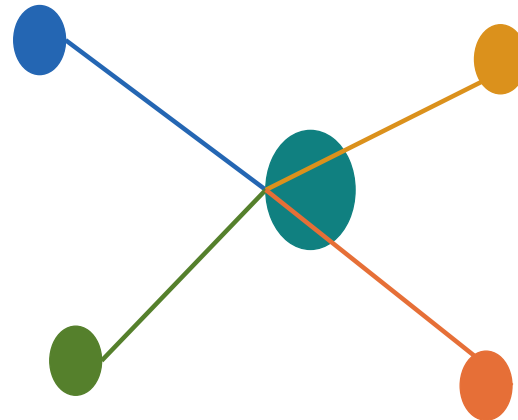
Keeps the strongest single path



$$A(v) = \max(\text{signal})$$

SUM activation

**Accumulates many weak paths
converging on the same answer node**



$$A(v) += \text{total incoming signal}$$

cycle guard 1: refractory

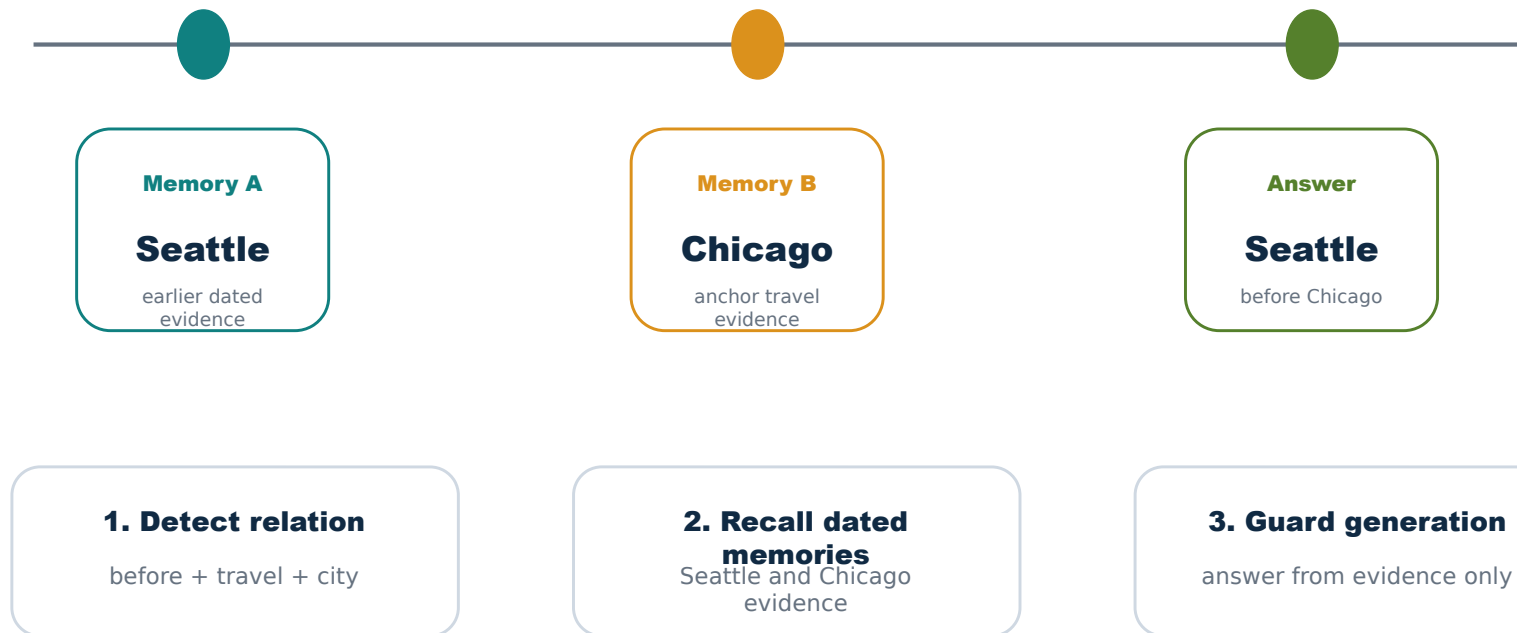
cycle guard 2: firing quota

cycle guard 3: saturation

Running example: before-travel question

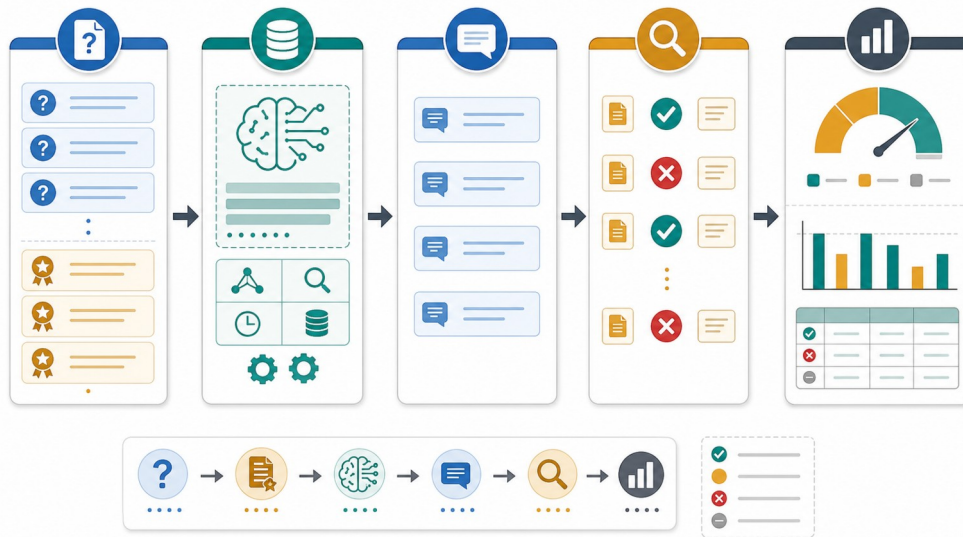
EXAMPLE

Query: Which city was the user in before traveling to Chicago?



Evaluation: judge PASS is the reported metric

EVALUATION



Reported scoring rules

- LLM-as-a-judge compares generated answer with gold.
- PASS/Total is the metric shown in this deck.
- Same judge policy across LongMemEval and LoCoMo.
- Internal run IDs and case IDs are hidden from reader-facing slides.

Question

CogMem
answer

Judge verdict

Category totals

LongMemEval: multi-channel typed graph is strongest

RESULTS

31/35

Multi-channel + 6 types

judge PASS = 88.6%

30/35

HINDSIGHT baseline

judge PASS = 85.7%

35

Evaluation set

LongMemEval questions

Multi-channel + 6 types



88.6%

HINDSIGHT baseline



85.7%

Ablations

no habit stays 31/35; no action-effect/intention drops to 29/35

Takeaway: CogMem improves over HINDSIGHT when typed graph memory is fused with lexical, semantic, and temporal evidence.

Control experiment: SUM improves graph-only top-k priority

RESULTS

0.8052

Graph-only SUM recall@5

mean session recall@5

0.7624

Graph-only MAX recall@5

mean session recall@5

+0.0429

Absolute lift

SUM minus MAX at top-5

Rank budget view



Relevant evidence should land inside ranks 1-5.

Result interpretation

- Recall@10 is equal at 0.8481.
- SUM better in 2/35 questions.
- MAX better in 0/35 questions.
- Top-10 order changes in 12/35 questions.

Honest claim: SUM does not create new evidence; it moves convergent evidence earlier in the prompt budget.

LoCoMo: judge PASS crosses the 70% target

RESULTS

119/161

**Final evidence-guard
configuration**

judge PASS = 73.9%

97/161

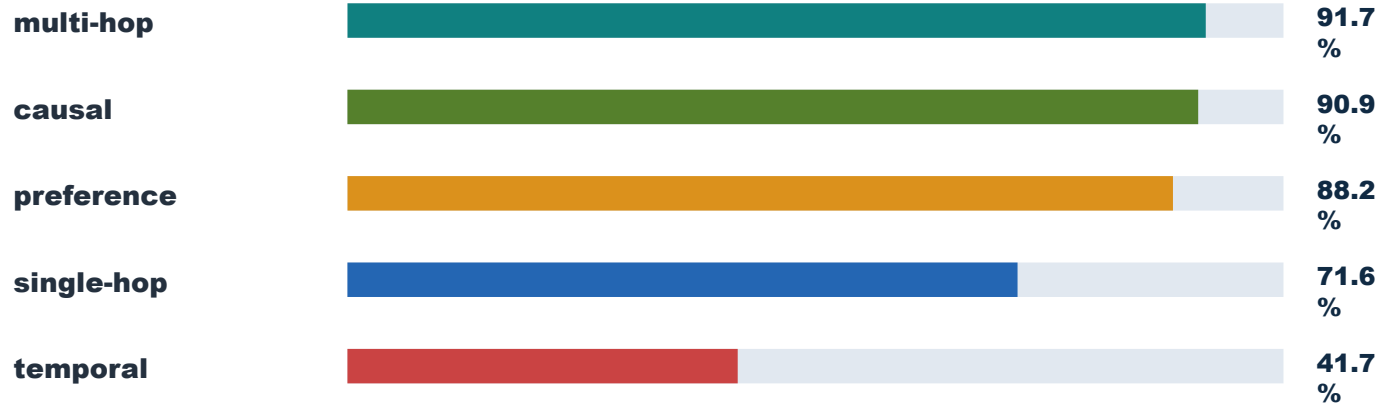
Previous baseline

judge PASS = 60.2%

+22

Recovered cases

additional PASS



Takeaway: the full system is useful on a harder long-dialogue benchmark, but temporal reasoning still needs focused work.

Qualitative proof: intention stores unfinished plans

TYPED MEMORY PROOF

Question: What sustainability habit did the user intend to start but has not?

Gold answer: Composting

FULL BANK: gold representation exists

37 facts | 4 intention nodes | cited [1,2,3,9,11]

[intention] plans to start composting



[intention] kitchen scraps -> reduce waste

[1] keeps meaning to compost, but has not gotten around to it yet

[9][11] compostable trash bags = supporting evidence, not the answer itself

Generated answer uses these cited nodes -> Composting

ABLATED BANK: gold representation removed

26 facts | 0 intention nodes | only decoy evidence remains

Missing component: no node says "plans to start composting"

[experience] rain barrel / rainwater garden

[5] started thinking about collecting rainwater; model answers the decoy topic

Discrimination claim: the gold answer exists only as typed intention facts in this case.

Qualitative proof: action-effect stores tool outcomes

TYPED MEMORY PROOF

Question: What does the agent do when Stripe returns HTTP 429 + Retry-After?

Gold: backoff, sleep, then 200

FULL BANK: causal triple is explicit

13 facts | 7 action_effect nodes | cited [1,2,3,9]

precondition:
HTTP 429 +
Retry-After
ignored



action:
respect
Retry-After +
exponential
backoff



outcome:
subsequent
calls return
200

[2] Added Retry-After handling eliminated 429s

[3] Naive retry without honoring Retry-After caused script exit

Generated answer can state both behavior and result.

ABLATED BANK: causal bridge is missing

12 facts | 0 action_effect nodes | recall is non-empty but incomplete

[1] Stripe API returns 429 and includes Retry-After header

[7][8] Backoff logic was added to the script

**Missing component: no typed link from condition
-> action -> 200 outcome**

Discrimination claim: action_effect stores the executable causal chain, not just adjacent facts.

What was proved, and what remains

CONTRIBUTION PROOF

Claim	Evidence	Status
Multi-channel memory recall	LongMemEval 31/35; LoCoMo 119/161	Strong
SUM graph activation	0.8052 vs 0.7624 recall@5	Strong, scoped
Evidence guard + snippets	+22 LoCoMo PASS cases	Strong practical
Intention / action-effect types	paired qualitative ablations: composting and HTTP 429	Conditional
Retain efficiency	~30 min vs 90-120; ~700 vs ~1500 nodes	Practical

Habit remains the least-proven type; future work should use plan lifecycle tasks, dense tool-action traces, and multi-week habit diaries.



HUST

THANK YOU !

Questions?

**Code, experiments, judge results, and
thesis artifacts are preserved in the
repository.**



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